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Lab Assignment:- (Group-A) 2

Code:-

```
#include <stdio.h>

void selectionSort(float arr[], int n)
{
    int i, j, min;
    float temp;
    for(i = 0; i < n - 1; i++){
        min = i;
        for(j = i + 1; j < n; j++){
            if(arr[j] < arr[min])
                min = j;}
        temp = arr[i];
        arr[i] = arr[min];
        arr[min] = temp;
    }
}

void bubbleSort(float arr[], int n)
{
    int i, j;
    float temp;

    for(i = 0; i < n - 1; i++){
        for(j = 0; j < n - i - 1; j++) {
            if(arr[j] > arr[j + 1])
             {
                temp = arr[j];
                arr[j] = arr[j + 1];
```

```
                arr[j + 1] = temp;
            }
        }
    }
}

void display(float arr[], int n)
{
    int i;

    printf("\nSorted Percentages:\n");
    for(i = 0; i < n; i++){
        printf("%.2f ", arr[i]);
    }
    printf("\n");
}

void topFive(float arr[], int n)
{
    int i, count;

    printf("\nTop Five Scores:\n");

    count = (n < 5) ? n : 5;

    for(i = n - 1; i >= n - count; i--)
    {
        printf("%.2f\n", arr[i]);
    }
}
```

```
int main()
{
    float per[100];
    int n, i, choice;

    printf("Enter number of students: ");
    scanf("%d", &n);

    printf("Enter first year percentages:\n");
    for(i = 0; i < n; i++){
        scanf("%f", &per[i]);
    }

    do{
        printf("\n===== MENU =====\n");
        printf("1. Selection Sort\n");
        printf("2. Bubble Sort\n");
        printf("3. Display Top Five Scores\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch(choice){
            case 1:
                selectionSort(per, n);
                printf("\nArray Sorted Using Selection Sort.\n");
                display(per, n);
                break;

            case 2:
                bubbleSort(per, n);
                printf("\nArray Sorted Using Bubble Sort.\n");
```

```
            display(per, n);
            break;

        case 3:
            topFive(per, n);
            break;

        case 4:
            printf("Exiting Program...\n");
            break;

        default:
            printf("Invalid Choice!\n");
    }

} while(choice != 4);

return 0;
}
```

Output:-

```
Enter number of students: 6
Enter first year percentages:
70
25
78
57
90
45

===== MENU =====
1. Selection Sort
2. Bubble Sort
3. Display Top Five Scores
4. Exit
Enter your choice: 1

Array Sorted Using Selection Sort.

Sorted Percentages:
25.00 45.00 57.00 70.00 78.00 90.00

===== MENU =====
1. Selection Sort
2. Bubble Sort
3. Display Top Five Scores
4. Exit
Enter your choice: 2

Array Sorted Using Bubble Sort.
```

```
Sorted Percentages:
25.00 45.00 57.00 70.00 78.00 90.00

===== MENU =====
1. Selection Sort
2. Bubble Sort
3. Display Top Five Scores
4. Exit
Enter your choice: 3

Top Five Scores:
90.00
78.00
70.00
57.00
45.00

===== MENU =====
1. Selection Sort
2. Bubble Sort
3. Display Top Five Scores
4. Exit
Enter your choice: 4
Exiting Program...
```